

Sicheng Lai

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Leo-Lsc

EDUCATION

The Hong Kong University of Science and Technology

Sept. 2026 – July 2027

M.Sc. in Big Data Technology

The Chinese University of Hong Kong, Shenzhen

Sept. 2022 – July 2026

B.Sc. in Computer Science and Engineering

Exchange Semester in University of California San Diego

March 2025 – July 2025

PUBLICATIONS

[†] Equal Contribution. ^{*} Corresponding author.

Yadi Cao[†], **Sicheng Lai**[†], Jiahe Huang[†], Yang Zhang[†], Zach Lawrence[†], Rohan Bhakta[†], Izzy F. Thomas[†], Mingyun Cao[†], Chung-Hao Tsai, Zihao Zhou, Yidong Zhao, Hao Liu, Alessandro Marinoni, Alexey Arefiev, and Rose Yu^{*}

“SimulCost: A Cost-Aware Benchmark for Automating Physics Simulations with LLMs”.

ICML 2026. Project Website.

Sicheng Lai[†], Dingjie Song[†], Mingxuan Wang, Shunian Chen, Lichao Sun, and Benyou Wang^{*}

“Both Text and Images Leaked! A Systematic Analysis of Data Contamination in Multimodal LLM”.

ICML 2025 DIG-BUGs Oral. EMNLP 2025 Findings.

RESEARCH EXPERIENCE

SimulCost: A Cost-Aware Playground

April 2025 - March 2026

UC San Diego | Co-First Author

Supervisor: Prof. [Rose Yu](#)

- Developed [SimulCost](#), a comprehensive benchmarking framework evaluating LLM cost-accuracy trade-off on scientific simulation optimization across 13 physics solvers and 40+ hyperparameter tuning tasks.
- Engineered a scalable Playground API enabling on-demand simulation generation, with caching mechanisms and pandas DataFrame-based data persistence reducing redundant computations by 50%.
- Benchmarked LLMs against brute-force (e.g., grid search) and Bayesian optimization (e.g., Gaussian process) baselines across 6 real-world scenarios, conducting statistical analysis and ablation studies.

Data Contamination in Multimodal LLMs

June 2024 - December 2024

CUHK-Shenzhen | Co-First Author

Supervisor: Prof. [Benyou Wang](#)

- Pioneered systematic multimodal data contamination detection by designing the [MM-Detect](#) framework, incorporating two novel methods: *Option Order Sensitivity Test* and *Slot Guessing for Perturbed Caption*.
- Validated the framework’s effectiveness and sensitivity by training LLaVA-1.5-7B variants with controlled leakage at early, mid, and late training stages, and varying contamination ratios (10%, 50%, 100%).
- Developed a heuristic method to trace contamination origins, revealing unimodal contamination in base LLMs and cross-modal contamination through training data overlap analysis.

INTERNSHIPS

Multimodal Algorithm Engineer
Tsinghua Shenzhen International Graduate School

December 2024 - March 2025
Shenzhen, China

- Optimized MLLM configurations through systematic experimentation with various combinations of LLMs (Qwen2.5/Vicuna) and visual encoders (CLIP/MLCD/SigLIP) based on the LLaVA-Next framework.
- Evaluated model performance using VLMEvalKit and lmms-eval to determine the optimal configuration (Qwen2.5 + SigLIP), achieving top 20% ranking on the OpenCompass MLLM Leaderboard.
- Trained the Qwen2.5 + SigLIP model on agricultural datasets curated and augmented with data-juicer, achieving expert-level agricultural capabilities while maintaining high general-domain performance.

PROJECTS

LLM Pretraining, Data Curation, Scaling, and Alignment
Stanford CS336

March 2026 - May 2026
Spring 2026

- Built a GPT-style language modeling stack from scratch in PyTorch, including byte-level BPE tokenization, RMSNorm/SwiGLU/RoPE Transformer blocks, AdamW, gradient clipping.
- Built TinyStories/OpenWebText pretraining pipelines with memory-mapped datasets, warmup + cosine LR decay, 10K/32K BPE tokenizers, 2.73B encoded OWT tokens, and best TinyStories val loss 1.3862.
- Reproduced Chinchilla-style IsoFLOPs scaling laws and built a local mini-scaling experiment with a custom PyTorch GPT, 15 runs over 584M tokens on 6×RTX 4090s, extrapolating to a held-out 3.3× larger compute budget with 0.7% validation-loss prediction error.
- Implemented a Common Crawl data filtering stack covering HTML extraction, fastText language/NSFW/toxicity/quality classifiers, PII masking, Gopher quality heuristics, exact line deduplication, and MinHash+LSH near-deduplication.
- Scaled web data curation from a single WET file to 50 Common Crawl WET shards, producing 83.8M clean tokens, then ran a controlled clean-vs-raw pretraining experiment showing filtered data reduced C4-100 perplexity from 108.54 to 89.77 (17.3%).
- Implemented GRPO and variants, including Dr.GRPO, RFT, MaxRL, clipping, GSPO sequence-level importance weighting, ZVP-GRPO token advantages, and gradient-accumulated microbatch training.
- Built a vLLM + HuggingFace RL training pipeline with NCCL weight sync, on/off-policy updates, and GSM8K evaluation; improved OLMo-2-1B reasoning accuracy from near-zero prompting to 42–45% standard GRPO validation accuracy and 51.4% best GSPO long-run accuracy.

HONORS

Outstanding Graduate (Exceptional achievements and contributions)
Dean’s List (Outstanding academic merit)
Undergraduate Research Awards (Key research contributions)

2026
2022 – 2025
2024 – 2025

TEACHING

Undergraduate Student Teaching Fellow (STA2001: Probability and Statistics I)

Spring 2024

CORE COMPETENCIES

- Programming & Systems:** Python, Java, C/C++ , RISC-V
- LLM:** PyTorch, Transformers, LangChain, Accelerate, DeepSpeed, bitsandbytes, vLLM, SGLang, Ollama
- Data Science:** NumPy, pandas, SciPy, Dask, xarray, Matplotlib, Seaborn, Weights & Biases
- Developer Tools & Deployment:** Hydra, Git, Conda, Poetry, uv, Docker, Kubernetes, SLURM, Linux
- Backend & Web:** Flask, REST APIs, SQLite, HTML/CSS/JavaScript, TypeScript, Astro, Tailwind CSS